

Do G7 Climate Commitments Match Reality?

Executive Summary

Project Title: Do G7 Climate Commitments Match Reality? A Data-Driven Analysis of Emissions & Targets (2015–2024)

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Problem Statement

The G7 countries – United States, United Kingdom, Germany, France, Italy, Canada, and Japan – have made ambitious climate pledges to reduce greenhouse gas emissions by 2030. However, there is growing concern that these commitments may not be backed by sufficient action. This project analyses emissions data from 2015 to 2024 to assess whether G7 countries are on track to meet their 2030 targets.

Data & Methods

- Data Sources: Our World in Data (based on Global Carbon Project) and official G7/UNFCCC submissions.
- Dataset: 7 G7 countries, emissions data (2015–2024), and 2030 target levels.
- Tools: Google Sheets (Tier 1), Power BI (Tier 2), Python (Tier 3).

Finding	Detail
UK Leads	The United Kingdom reduced emissions by 26.1% since 2015 – the best performance among G7 countries.
Canada Lags	Canada is the only G7 country with rising

	emissions (+6.5%) since 2015.
No Country on Track	Every G7 country remains above its 2030 target. No country is currently on track based on 2024 estimates.
US Has Largest Gap	The United States needs to reduce 1,780 MtCO ₂ more to meet its 2030 target – the largest gap.
France Has Smallest Gap	France needs only 120 MtCO ₂ more – the smallest gap among G7 countries.
Total Gap	The total emissions reduction needed across all G7 countries is 3,475 MtCO ₂ .

Visual Highlights

- Emissions Trend (2015–2024): All G7 countries show a downward trend, but Canada's emissions increased.
- Gap to Target (2024): The US has the largest gap; France has the smallest.
- 2024 vs Target: Every country is still above its 2030 target.
- Clustering Analysis: Three groups emerge – "Critical Need" (US, Japan), "Moderate" (Germany, Canada), and "On Track" (UK, France, Italy).

Recommendations

1. Prioritise Canada – as the only country with rising emissions, it needs the most urgent intervention.
2. Support the US – with the largest absolute gap, the US requires aggressive policy action.
3. Celebrate the UK – the UK's progress offers a model for others to follow.
4. Increase ambition – no country is on track; all need to accelerate their efforts.
 - End-to-end data analysis – from data collection to insights.
 - Multi-tool proficiency – Google Sheets, Power BI, Python.
 - Policy storytelling – connecting data to real-world decisions.
 - Statistical rigour – correlation, regression, clustering, t-tests.

- **Conclusion**

The G7's climate pledges are ambitious, but the data shows a significant gap between promises and progress. No country is currently on track to meet its 2030 target. While the UK leads in emissions reduction, Canada is falling behind. Urgent action is needed across all G7 nations to translate commitments into measurable outcomes.